COMP3013 2024 Fall

Assignment 3

For the submission, please pack all files and convert them into **A SINGLE PDF FILE**. Rename the PDF file as COMP3013_24F_A3_XXXX, where XXXX is your student ID.

The schema of a database for public transportation companies is as follows. Keys are underlined.

- $company = (\underline{cID}, cname, address, phone)$
- $route = (\underline{rID}, departure, arrival, cID)$
  // cID is a foreign key to $company.cID$.
- $vehicle = (\underline{plateNum}, model, capacity, manufacturer, cID)$
  // cID is a foreign key to $company.cID$.
- $driver = (\underline{dID}, name, gender, age, cID)$
  // cID is a foreign key to $company.cID$.
- $serve = (\underline{rID, plateNum})$
  // rID is a foreign key to $route.rID$.
  // $plateNum$ is a foreign key to $vehicle.plateNum$.
- $drive = (\underline{plateNum}, dID)$
  // dID is a foreign key to $diver.dID$.
  // $plateNum$ is a foreign key to $vehicle.plateNum$.

Q1. Write a query for each following question. (10pt for each)

a) Find the ID of routes which are not served by any vehicle. You must use subqueries.
b) Find the name of drivers who have served the route 69 (rID). You must use subqueries.
c) Find the name of drivers who have driven all vehicles.
d) Find the plate number of vehicles which have served all route operated by "Xinhe" (company name).
e) Implement constraints to guarantee the gender of a driver is either "Male" or "Female" and the age is from 20 to 60.

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-- a) Find the ID of routes which are not served by any vehicle.
You must use subqueries.
SELECT rID
FROM route
WHERE rID NOT IN (
        SELECT rID
        FROM serve
    );

-- b) Find the name of drivers who have served the route 69 (rID).
You must use subqueries.
SELECT name
FROM driver
WHERE dID IN (
        SELECT dID
        FROM driver
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        WHERE plateNum IN (
                SELECT plateNum
                FROM serve
                WHERE rID = 69
            )
    );

-- c) Find the name of drivers who have driven all vehicles.
SELECT name
FROM driver
WHERE dID IN (
        SELECT dID
        FROM drive
        GROUP BY dID
        HAVING COUNT(DISTINCT plateNum) = (
                SELECT COUNT(*)
                FROM vehicle
            )
    );

-- d) Find the plate number of vehicles which have served all
route operated by “Xinhe” (company name).
SELECT plateNum
FROM vehicle
WHERE plateNum IN (
        SELECT plateNum
        FROM route
        WHERE rID IN (
                SELECT rID
                FROM route
                WHERE cID = (
                        SELECT cID
                        FROM company
                        WHERE cname = "Xinhe"
                    )
            )
        GROUP BY plateNum
        HAVING COUNT(DISTINCT rID) = (
                SELECT COUNT(*)
                FROM route
                WHERE cID = (
                        SELECT cID
                        FROM company
                        WHERE cname = 'Xinhe'
                    )
            )
    );

-- e) Implement constraints to guarantee the gender of a driver
is either “Male” or “Female” and the age is from 20 to 60.
ALTER TABLE driver
ADD CONSTRAINT chk_gender CHECK (gender IN ('Male', 'Female')),
    ADD CONSTRAINT chk_age CHECK (
        age BETWEEN 20 AND 60
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    );
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Q2. Given an instance of a relational schema $R = \{A, B, C\}$ and a list of functional dependencies.

A	B	C
1	2	2
1	3	2
1	4	2
2	5	2

Decide whether the functional dependency is satisfied by the instance. (10 pt)

a) $A \to B$ b) $A \to C$ c) $B \to A$

d) $B \to C$ e) $C \to A$ f) $C \to B$

g) $AB \to C$ h) $AC \to B$ i) $BC \to A$

a)	$A \to B$	when A = 1, B = {2, 3, 4}, so (a) is **not satisfied**.
b)	$A \to C$	when A = 1, C = 2; when A = 2, C = 2; so (b) is **satisfied**.
c)	$B \to A$	when B = 2, 3, 4, A = 1; when B = 5, A = 2; so (c) is **satisfied**.
d)	$B \to C$	B = 2, 3, 4, 5, C = 2; so (d) is **satisfied**.
e)	$C \to A$	C = 2, A = {1, 2}, so (e) is **not satisfied**.
f)	$C \to B$	C = 2, B = {2, 3, 4, 5}, so (f) is **not satisfied**.
g)	$AB \to C$	AB = {(1, 2), (1, 3), (1, 4), (2, 5)}, C = 2, so (g) is **satisfied**.
h)	$AC \to B$	when AC = (1, 2), B = {2, 3, 4 ,5}, so (h) is **not satisfied**.
i)	$BC \to A$	BC = {(2, 2), (3, 2), (4, 2), (5, 2)}, A = {1, 2}, (i) is **satisfied**.

Q3. Given a relational schema and a set of functional dependencies

- $R = \{A, B, C, D, E\}$
- $F = \{AB \to CD, BC \to DE, CD \to E, DE \to A\}$

a) Find all candidate keys of R. (6 pt)
b) Decompose R into BCNF. Show the steps. (15 pt)
c) Does the BCNF decomposition in part b) preserve all functional dependencies? (4 pt)
d) Decompose R into 3NF. Show the steps. (15 pt)

a)

$AB^{+} = \{\mathrm{A, B, C, D, E}\}$, $\mathrm{BC}^{+} = \{\mathrm{A, B, C, D, E}\}$

Candidate keys are AB and BC.

b)

Firstly, we know CD and DE is not super key, which violates BCNF.

Thus we need to decompose R.

For the dependency $CD \to E$, we decompose R into:

$R_1 = \{C, D, E\}, F_1 = \{CD \to E\}$

$$R_2 = \{A, B, C, D\}, F_2 = \{AB \to CD, BC \to DE, DE \to A\}$$

For the dependency $DE \to A$, we decompose R_2 into:

$$R_3 = \{D, E, A\}, F_3 = \{DE \to A\}$$

$$R_4 = \{B, C, D\}, F_4 = \{AB \to CD, BC \to DE\}$$

Thus, the BCNF decomposition results in the following relations:

$R_1 = \{C, D, E\}$

$R_3 = \{\mathrm{A}, D, E\}$

$R_4 = \{B, C, D\}$

c)

1. $AB \to CD$

   $AB \to C$ lost, $BC \to D$ in R_4

2. $BC \to DE$

   $BC \to D$ in R_4, $CD \to E$ in R_1

3. $CD \to E$ in R_1
4. $DE \to A$ in R_3

The BCNF decomposition does not preserve all functional dependencies, specifically $AB \to C$

d)

$$\mathrm{F} = \{AB \to CD, BC \to DE, CD \to E, DE \to A\}$$

1. Convert FDs to Minimal Cover:

$F_c = \{AB \to CD, BC \to DE, CD \to E, DE \to A\}$

D is extraneous in $AB \to CD$

E is extraneous in BC → DE

$F_c = \{\mathrm{AB} \to C, \mathrm{BC} \to D, \mathrm{CD} \to E, DE \to A\}$

2. Create a Relation for Each Functional Dependency in the Minimal Cover:

$R_1 = \{A, B, C\}$

$R_2 = \{\mathrm{B, C, D}\}$

$R_3 = \{\mathrm{C, D, E}\}$

$R_4 = \{\mathrm{A, D, E}\}$

3. R_1 already contains a candidate key (AB) for R

4. Return $\{\ \{A, B, C\}, \{B, C, D\}, \{C, D, E\}, \{A, D, E\}\ \}$